MIL-C-47068(MI)
10 May 1974
SUPERSEDING
MIS 14209
1 March 1965

MILITARY SPECIFICATION

COATING, CONDUCTIVE, EPOXY, SILVER

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a highly conductive, epoxy base silver coating material used for electronic shielding purposes.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-781	Ethylene Glycol Monoethyl Ether, Technical
TT-M-268	Methyl Isobutyl Ketone (for use in Organic Coatings)
TT-T-548	Toluene, Technical
PPP-C-96	Cans, Metal, 28 Gage and lighter
PPP-C-200	Chemicals, Liquid, Packaging, and Packing of

FSC 8010

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STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing
FED-STD-406	Plastics, Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Publications

Scott's Standard Method of Chemical Analysis	Gay-Lussac Method for Sliver Bullion

(Application for copies should be addressed to the D. Van Nostrand Company, Princeton, New Jersey 08540.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

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(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington, D.C. 20013.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample (see 4.2) of material shall meet the requirements of this specification.

3.2 Material. The material shall consist of an epoxy resin, pigmented with pure silver powder and an aliphatic amine type catalyst. The material, when mixed, may be reduced with a thinner (see 3.2.1) for spraying purposes. The material shall be capable of being cured at a temperature not exceeding 71 degrees Celsius (C) (160 degrees Fahrenheit (F)).

3.2.1 Thinner. The composition of the thinner shall conform to the requirements specified in Table I.

Table I

Thinner composition		
Ingredient	Specification	Parts by volume
Ethylene Glycol Monoethyl Ether	TT-E-781	1
Methyl Isobutyl Ketone	TT-M-268	1
Toluene	TT-T-548	1

3.3 Electrical property.

3.3.1 Surface resistivity. The surface resistivity of the cured coating shall be less than 0.4 ohms.

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3.4 Physical properties.

3.4.1 Density. The density of the cured coating shall be 2.21 to 2.37 grams per cubic centimeter (cc) at 23.9 plus or minus 1.1 degrees Celsius (C) (75 plus or minus 2 degrees Fahrenheit (F)).

3.4.2 Silver content. The silver content of the epoxy base resin shall be 67 percent minimum, by weight.

3.4.3 Specific gravity. The specific gravity of the thinner shall be 0.867 plus or minus 0.004 at 23.9 plus or minus 1.1 degrees C (75 plus or minus 2 degrees F).

3.4.4 Suspension of solids. The epoxy base resin shall show no more than slight settling, no caking and shall redisperse to a smooth homogeneous state.

3.4.5 Shelf life. The shelf life of the epoxy base resin and catalyst shall not be less than 12 and 6 months respectively from the date of manufacture when stored in original unopened containers at temperature less than 26.7 degrees C (80 degrees F).

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform, in conformance with this specification, and properly packaged and packed. The material shall be free of dirt, foreign material or other contamination. Containers shall be clean, uniformly filled, well sealed, legibly and durably marked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified (see 6.2),

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the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Classification of examinations and tests. The examination and testing of material shall be classified as follows:

a. Preproduction tests (4.4).

b. Quality conformance tests (4.5).

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examinations and tests:

a. Workmanship (see 3.5).

b. Surface resistivity (see 4.5.1.1).

c. Specific gravity (see 4.6.2.3), when thinner is procured as a mixed formulation.

d. Suspension of solids (see 4.6.2.4).

4.5.1 Lot size and sampling.

4.5.1.1 Lot size. Lot size shall consist of all of the epoxy base resin, catalyst and thinner manufactured at one time from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

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4.5.1.2 Sampling. Unless otherwise specified (see 6.2), one pint of epoxy base resin, 1/8 pint of catalyst and 1/2 pint of thinner from one unit, representative of each lot (see 4.5.1.1) shall be selected at random for quality conformance testing. Each container of pigmented resin, catalyst and thinner shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.6 Test methods.

4.6.1 Electrical property test.

4.6.1.1 Surface resistivity.

4.6.1.1.1 Test specimen preparation. A coating of material, thoroughly mixed, and thinned shall be sprayed onto one side of a standard microscope glass slide, approximately 1 inch wide by 3 inches long. Cure coating at a temperature not exceeding 71 degrees C (160 degrees F). Apply additional coats of material in the same manner to obtain a coating thickness not greater than 0.003 inch.

4.6.1.1.2 Test procedure. The surface resistivity shall be determined in accordance with the following procedure. The contacts shall be placed on the prepared test surface of the slide (see 4.6.1.1.1) approximately 1.5 inches apart. The surface resistance shall be measured with a volt ohm-milliameter to determine compliance with requirements of 3.3.1.

4.6.2 Physical property tests.

4.6.2.1 Density. The density specified in 3.4.1 shall be determined in accordance with FED-STD-406, Method 5011.

4.6.2.2 Silver content. The silver content specified in 3.4.2 shall be determined in accordance with Scott's Standard Method of Chemical Analysis, Gay-Lussac Method for Silver Bullion.

4.6.2.3 Specific gravity. The specific gravity specified in 3.4.3 shall be determined in accordance with FED-STD-141, Method 4183.

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4.6.2.4 Suspension of solids. Compliance with the suspension of solids requirement specified in 3.4.4 shall be determined by stirring the contents of the container and visually examining for uniformity of dispersion and evidence of caking. Caking is defined as material which has settled on the bottom of the container and cannot be readily mixed by stirring.

4.6.2.5 Shelf life. The epoxy base resin and catalyst shall be tested for conformance with all the requirements specified herein after completion of storage period. The manufacturer's certificate of compliance with 3.4.5 may be accepted during the storage period.

4.7 Preservation, packaging, packing and marking. The preservation, packaging, packing and marking shall be examined for compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A. The epoxy base resin and thinner shall be packaged in either cans or bottles of the size specified (see 6.2). The catalyst shall be packaged in glass bottles. The cans shall conform to PPP-C-96, Type V, Class 4, and the cans shall be packaged in accordance with the Level A requirements of the appendix to PPP-C-96. Bottles shall be packaged in accordance with Level A requirements of PPP-C-300.

5.1.2 Level C. Material shall be packaged to provide adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Material packaged in cans as specified in 5.1.1 shall be packed in accordance with Level A requirements of the appendix to PPP-C-96 except that the exterior shipping containers shall conform to the requirement of PPP-B-601 overseas Type, Style I. Material packaged in bottles as specified in 5.1.1 shall be packed in accordance with the Level A requirements of PPP-C-300.

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5.2.2 Level B. Material packaged in cans or bottles as specified in 5.1.1 shall be packed in accordance with Level A requirements of the appendix to PPP-C-96 or PPP-C-300 as applicable.

5.2.3 Level C. Material shall be packed in such a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The silver epoxy conductive coating material is intended for use as electronic shielding for welded modules.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1).

c. Where preproduction sample test shall be performed (see 4.2).

d. Sampling plan if other than specified (see 4.5.1.2).

e. Any special marking requirements (see 5.2).

f. Container size (see 5.1).

g. Selection of applicable levels of preservation, packaging, and packing.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14209, dated 1 March 1965.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 8010-A094

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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